

Study and mitigation of dc bus voltage imbalance in three level neutral point clamped inverters

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Abstract—This document, written as part of a mini project, talks about the relevance of the DC bus voltage imbalance problem and potential fixes for the same.

I. INTRODUCTION

An inverter is an electronic device that converts direct current (DC) to alternating current (AC).

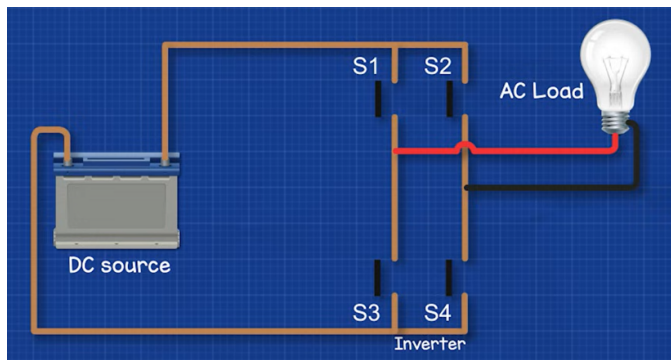


Fig. 1: Generalized function of an inverter

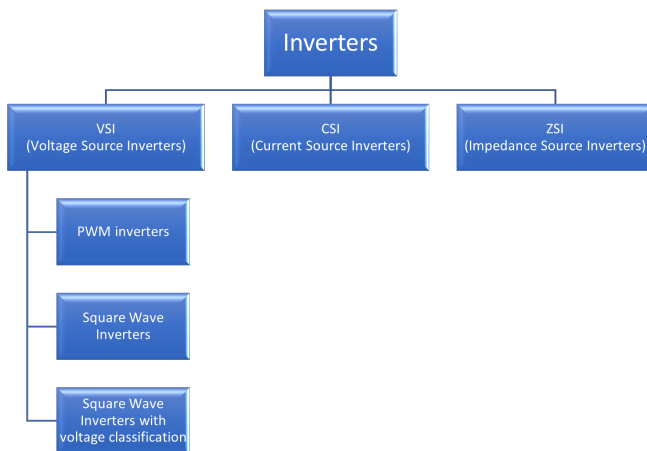


Fig. 2: Classifications of inverters

A. Applications of inverters

- UPS systems – Energy stored in a battery needs to be able to power AC appliances.
- Variable frequency drives – Inverters provide a simple and convenient way of controlling both the output voltage

and the frequency, enabling motors to operate at optimal efficiency under different load conditions.

- Renewable energy systems – The generated power almost always needs an inverter to convert it to be used by household appliances.
- Power factor quality – Inverters can produce reactive power as needed to maintain an optimal power factor, thus reducing transmission losses.

II. 2 LEVEL INVERTERS

A 2-level inverter is the simplest and most common type of voltage-source inverter. It gets its name because it can only produce two distinct voltage levels ($+V_{dc}/2$ and $-V_{dc}/2$) at the output of a leg at a given instant.

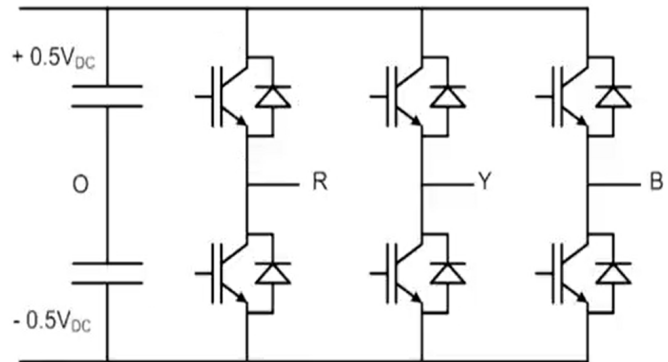


Fig. 3: Circuit diagram of a 3-phase, 2-level inverter

A. Output at a single leg

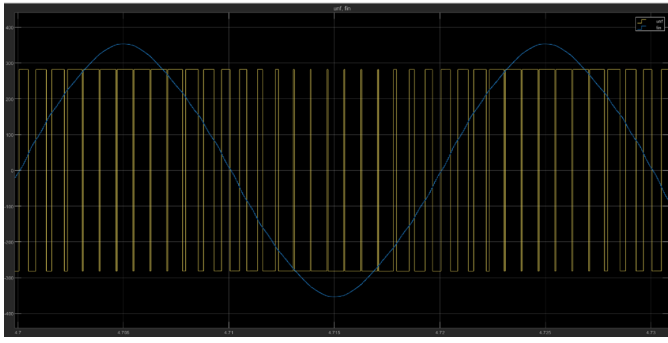
Apart from the fundamental component, we see harmonics of the switching frequency and their sidebands, as seen in Fig.4.

B. Single phase topology

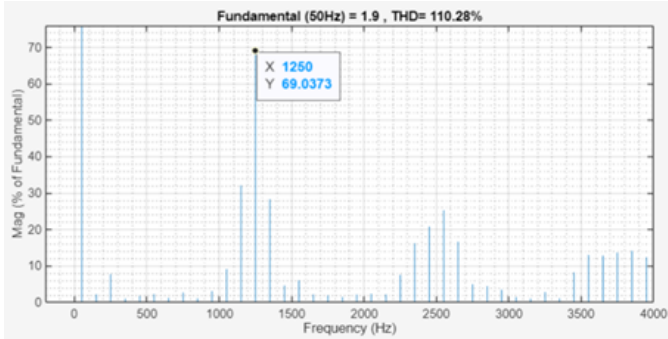
In Fig.5, we see that the switching frequency component is absent from the FFT. Since the component had the same phase in both legs, it gets canceled out.

C. Three phase topology

Three phase shows the same cancellation of the switching frequency component and results in a waveform similar to that of the single phase inverter (Fig.6).

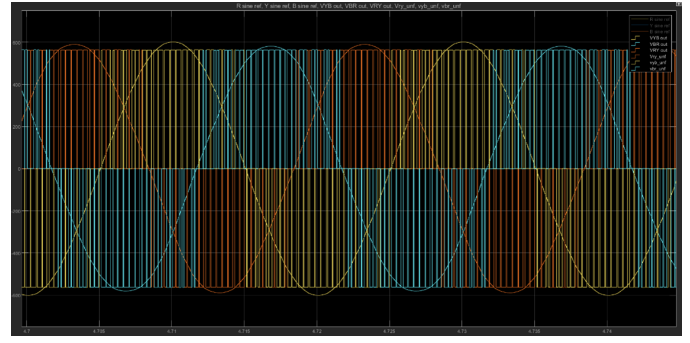


(a) Pulsed and filtered waveforms

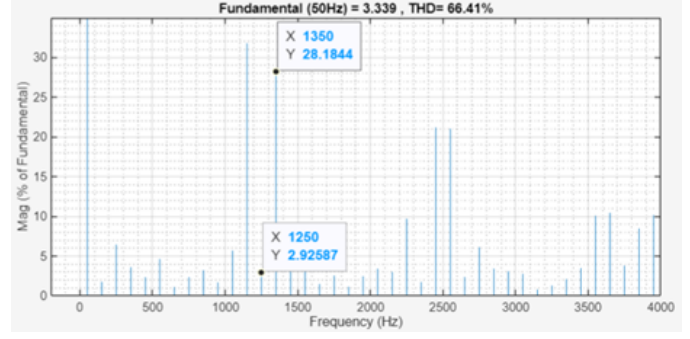


(b) FFT of the pulsed waveform

Fig. 4: Output at a leg of a 2-level inverter

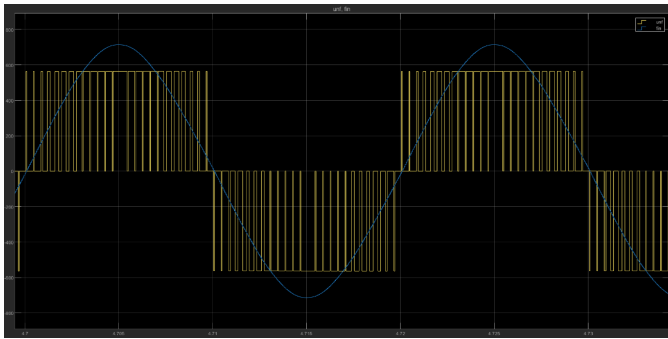


(a) Pulsed and filtered waveforms

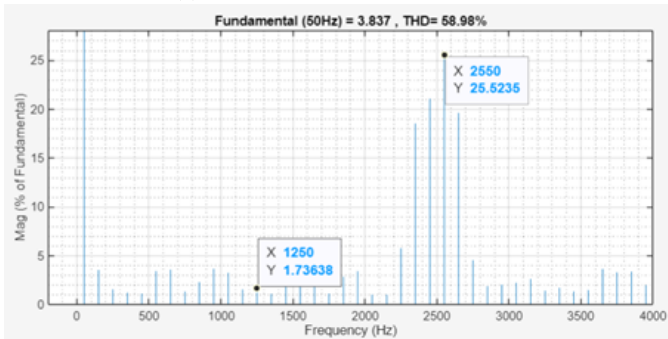


(b) FFT of the pulsed waveform

Fig. 6: Output of a 2-level three phase inverter



(a) Pulsed and filtered waveforms



(b) FFT of the pulsed waveform

Fig. 5: Output of a 2-level single phase inverter

III. 3 LEVEL INVERTERS

A 3-level inverter is an inverter designed to produce an output AC waveform with three distinct voltage levels ($+V_{dc}/2$,

0 and $+V_{dc}/2$) instead of two at any given instant.

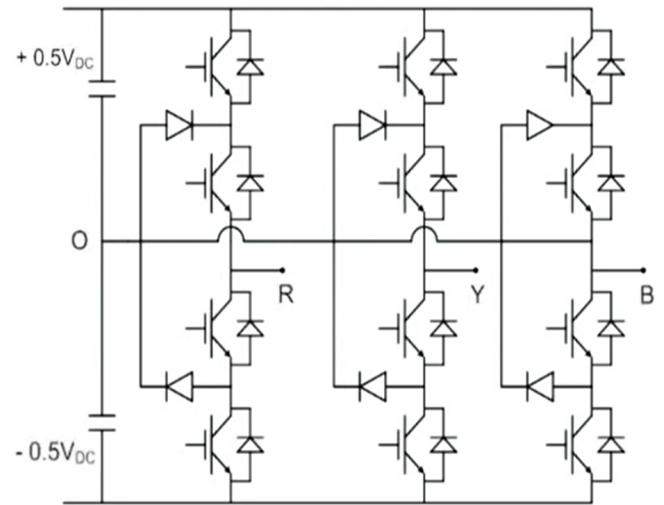


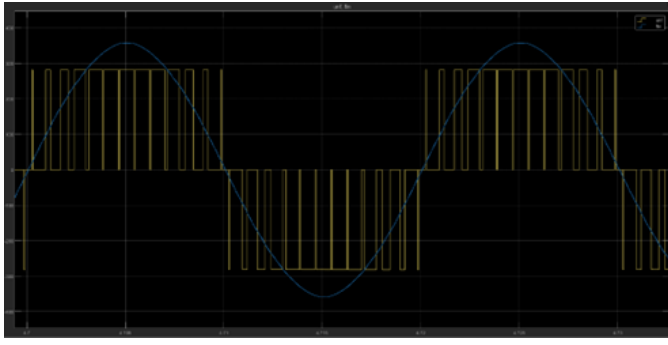
Fig. 7: Circuit diagram of a 3-phase, 3-level inverter

A. Output at a single leg

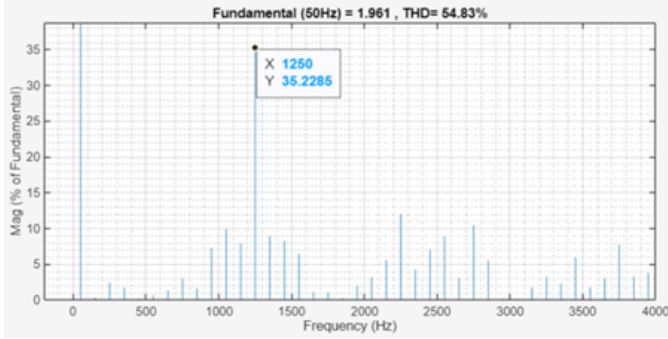
We see from Fig.8 that the added option of a 3rd level has made the leg voltage look significantly closer to the desired fundamental sine wave, thereby also improving the THD.

B. Single phase topology

Similar to the 2-level inverter, the switching frequency component is cancelled out, as seen in Fig.9.

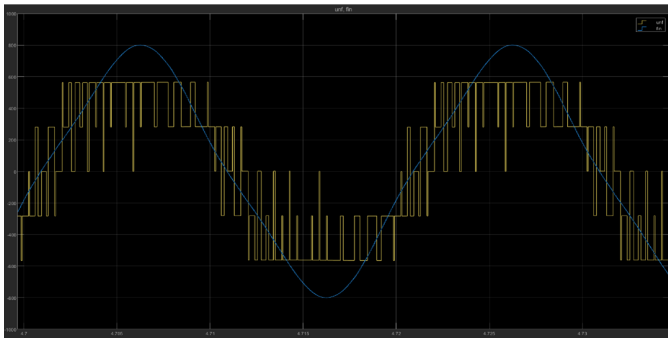


(a) Pulsed and filtered waveforms

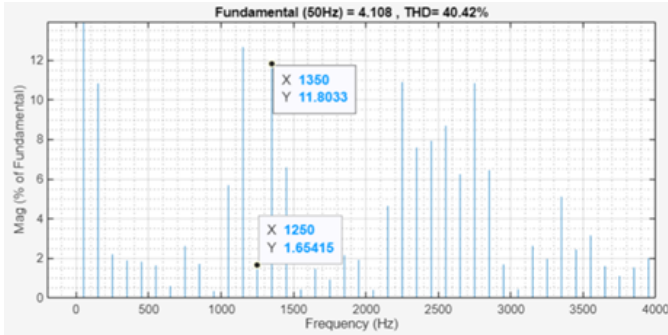


(b) FFT of the pulsed waveform

Fig. 8: Output at a leg of a 3-level inverter



(a) Pulsed and filtered waveforms



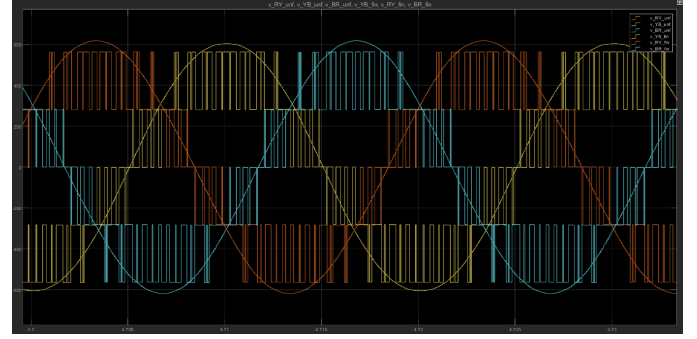
(b) FFT of the pulsed waveform

Fig. 9: Output of a 3-level single phase inverter

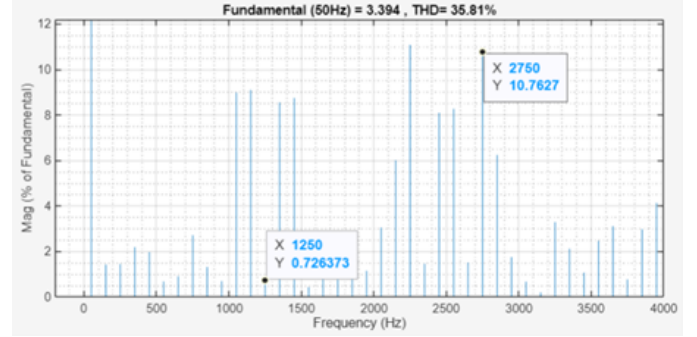
C. Three phase topology

Three phase shows the same cancellation of the switching frequency component and results in a waveform similar to that

of the single phase inverter (Fig.10).



(a) Pulsed and filtered waveforms



(b) FFT of the pulsed waveform

Fig. 10: Output of a 3-level three phase inverter

IV. ADVANTAGES OF 3-LEVEL INVERTERS

- The terminal voltage waveform of the 3-level NPC inverter shows less harmonic content than that of the conventional 2-level inverter [1].
- In a 3-level inverter, each device needs to block only half the voltage as in a 2-level inverter, thus reducing the device's cost and the switching loss of each device.
- They generate a smaller common-mode (CM) voltage, thus reducing the stress in the motor bearings [2].
- They draw input current with very low distortion and the output voltage has extremely low dv/dt [2].

V. CAPACITOR VOLTAGE IMBALANCE PROBLEM

Under certain conditions, the capacitor voltages stray away from $V_{dc}/2$, affecting the performance of the inverter negatively.

A. Causes

In practical circuits, various reasons such as unequal capacitance leakage currents, asymmetric current tracking, unequal delays in semiconductor devices, etc., could cause the capacitor voltages to drift away from the reference values, degrading the performance of the inverter [3].

However, in ideal circuits, the presence of dc components in the neutral current is the main reason. In a 2-level inverter, the source responsible for keeping the capacitors charged

provides any DC current required by the load. But, in a 3-level inverter, the DC current is able to flow through the capacitors themselves, through the connection at the DC bus midpoint. The injection of a dc component of current through the capacitors has the effect of charging one of the capacitors above its reference value and discharging the other below the reference value.

B. Problems caused by the imbalance

- Harmonic distortion is introduced in the output voltage [4].
- An excessive high voltage may be applied to some switching devices [5].

C. Dependence of the neutral point voltage on the neutral point current

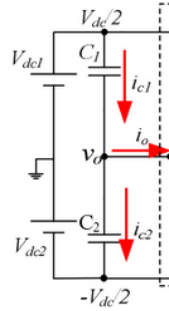


Fig. 11: Dependence of neutral point voltage on neutral point current

Assuming $C_1 = C_2 = C$, we have

$$i_{c1} = C \frac{d}{dt} \left(\frac{V_{dc}}{2} - v_o \right)$$

$$i_{c2} = C \frac{d}{dt} \left(\frac{V_{dc}}{2} + v_o \right)$$

Thus, we have

$$i_o = i_{c1} - i_{c2} = -2C \frac{d}{dt} (v_o)$$

$$v_o(t) - v_o(0) = -\frac{1}{2C} \int_0^t i_o(\tau) d\tau \quad [4] \quad (1)$$

This clearly shows us that if a constant imbalance is maintained over an extended period of time (as against having the imbalance for just over a single time period), the neutral-point current has a DC component.

We then write

$$i_o = D_{a,0} i_a + D_{b,0} i_b + D_{c,0} i_c \quad (2)$$

where $D_{a,0}$, $D_{b,0}$ and $D_{c,0}$ are the fractions of time for which the respective phase has 0 voltage; and i_a , i_b and i_c are the currents drawn by the respective phases.

For SPWM with 2 carriers (Fig.12), we can write the duty ratio of the neutral point as

$$D_0 = 1 + \frac{|v_c|}{\hat{V}_m}$$

Assuming $\hat{V}_m = 1$, D_o for each phase would be

$$D_{a,0} = 1 + |m_a \sin(\omega t)|$$

$$D_{b,0} = 1 + |m_a \sin(\omega t - 120^\circ)|$$

$$D_{c,0} = 1 + |m_a \sin(\omega t + 120^\circ)|$$

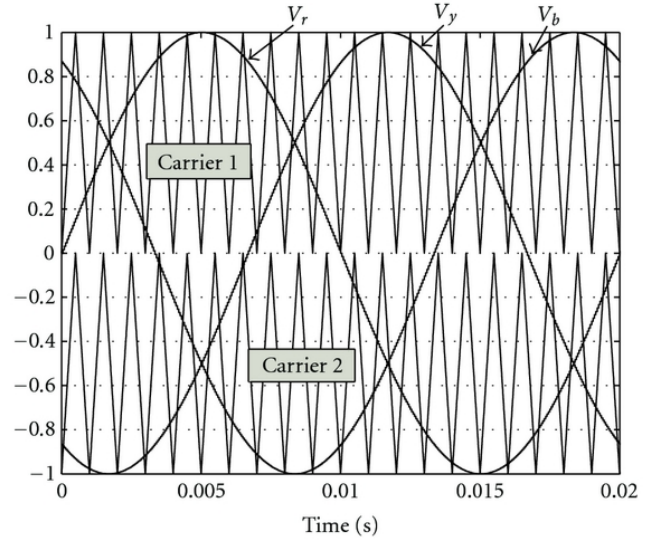


Fig. 12: Typical SPWM for a 3-level NCP inverter

VI. IMBALANCE PROBLEM IN A 3-WIRE SYSTEM

We first consider a 3-phase, balanced 3-wire load (neutral point of load is left unconnected) as in Fig.13.

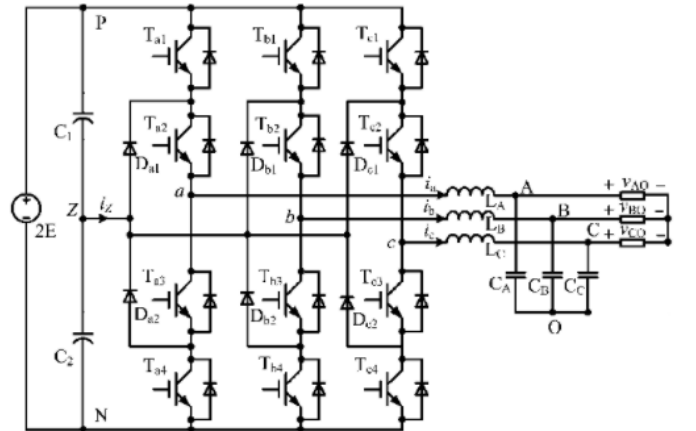


Fig. 13: 3-level NPC inverter with a 3-wire load

We can write

$$i_a = I_m \sin(\omega t - \phi)$$

$$i_b = I_m \sin(\omega t - \phi - 120^\circ)$$

$$i_c = I_m \sin(\omega t - \phi + 120^\circ)$$

Substituting the duty ratios and currents into equation (2) for $0^\circ \leq \omega t < 60^\circ$ (where the R and B phases' modulating waves are positive, but the Y phase's is negative), we have

$$\begin{aligned} i_o &= (1 - m_a \sin(\omega t)) I_m \sin(\omega t - \phi) \\ &+ (1 + m_a \sin(\omega t - 120^\circ)) I_m \sin(\omega t - \phi - 120^\circ) \\ &+ (1 - m_a \sin(\omega t - 120^\circ)) I_m \sin(\omega t - \phi + 120^\circ) \end{aligned}$$

$$\begin{aligned} i_o &= \frac{m_a I_m}{2} [\cos(2\omega t - \phi) - \cos \phi] \\ &+ \frac{m_a I_m}{2} [\cos \phi - \cos(2\omega t - \phi - 240^\circ)] \\ &+ \frac{m_a I_m}{2} [-\cos \phi + \cos(2\omega t - \phi + 240^\circ)] \end{aligned}$$

$$i_o = \frac{m_a I_m}{2} (-\cos \phi - 2 \cos(2\omega t - \phi - 240^\circ))$$

Similarly, we get

$$\begin{aligned} i_o &= \frac{m_a I_m}{2} (\cos \phi + 2 \cos(2\omega t - \phi)) \\ &; 60^\circ \leq \omega t < 120^\circ \end{aligned}$$

$$\begin{aligned} i_o &= \frac{m_a I_m}{2} (-\cos \phi - 2 \cos(2\omega t - \phi + 240^\circ)) \\ &; 120^\circ \leq \omega t < 180^\circ \end{aligned}$$

$$\begin{aligned} i_o &= \frac{m_a I_m}{2} (\cos \phi + 2 \cos(2\omega t - \phi - 240^\circ)) \\ &; 180^\circ \leq \omega t < 240^\circ \end{aligned}$$

$$\begin{aligned} i_o &= \frac{m_a I_m}{2} (-\cos \phi - 2 \cos(2\omega t - \phi)) \\ &; 240^\circ \leq \omega t < 300^\circ \end{aligned}$$

$$\begin{aligned} i_o &= \frac{m_a I_m}{2} (\cos \phi + 2 \cos(2\omega t - \phi + 240^\circ)) \\ &; 300^\circ \leq \omega t < 360^\circ \end{aligned}$$

We clearly see that i_o , when integrated (equation (1)) over a fundamental cycle, disappears. This implies that the unbalance returns to its initial state at the end of every fundamental cycle.

This implies that if the capacitor voltages are initially the same, the unbalance does not last for more than a switching cycle, as seen in Fig.14.

A similar argument can be made for unbalanced 3-phase loads, using the symmetrical component theory. There is no path for the zero sequence component to flow (due to the absence of a 4th wire) and the overall effect of both the positive and negative sequence components vanish over a switching cycle.

Thus, we conclude that the imbalance problem cannot occur in any 3-wire load systems under ideal conditions.

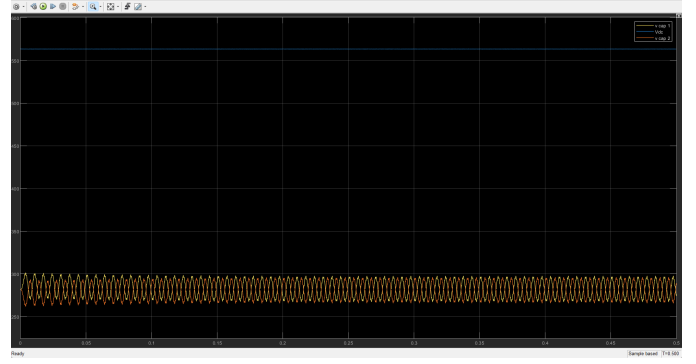


Fig. 14: DC bus capacitor voltages for a 3-wire RL load ($R = 10\Omega$, $L = 10mH$, $f_0 = 100Hz$, $L_{\text{filter}} = 20mH$, $R_{\text{filter}} = 10m\Omega$, $C_{\text{filter}} = 40\mu C$)

A. Observations regarding the periodic imbalance

We observe from the above equations that the mid-point current is proportional to the modulation index, m_a and the amplitude of load current, I_m . This means that for a 3-phase 3-wire RL load, smaller values of inductance or resistance could cause a very high voltage across one of the capacitors momentarily, which would then appear across the switches.

This would imply that the switches need to be selected based on this peak in the voltage, in the case where no balancing strategy is employed.

B. 4-wire load systems

The introduction of zero sequence currents in the load due to the 4th wire (Fig. 15) could directly result in a DC current through the neutral point. The neutral point voltage would then stray away from the midpoint of the DC bus, causing the capacitor voltages to become unequal.

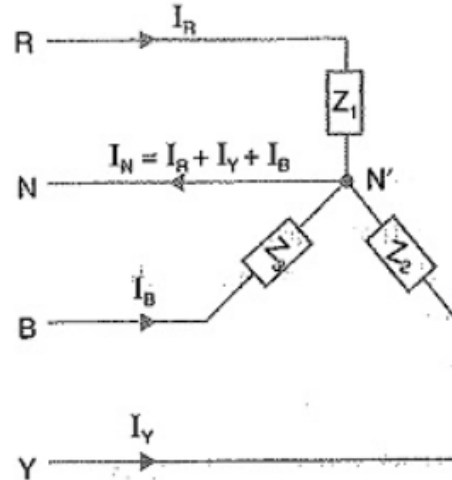


Fig. 15: A general 3-phase, 4-wire load

This kind of imbalance is more difficult to fix, especially using techniques involving modulating strategies, since they can only handle minor imbalances.

VII. SIMULATING THE VOLTAGE IMBALANCE

Simulation softwares are catered towards mimicking ideal systems, making it difficult to introduce the practical parameters that result in the imbalance. Despite this, there are still ways to force the unbalance to test the hypotheses and balancing strategies.

A. Initial voltages on the capacitors

In the absence of stray resistive elements (like snubbers), adding an initial (unbalanced) voltage to the simulation successfully maintains the imbalance even in the presence of a 3-wire (highly inductive) load (See Fig.16).

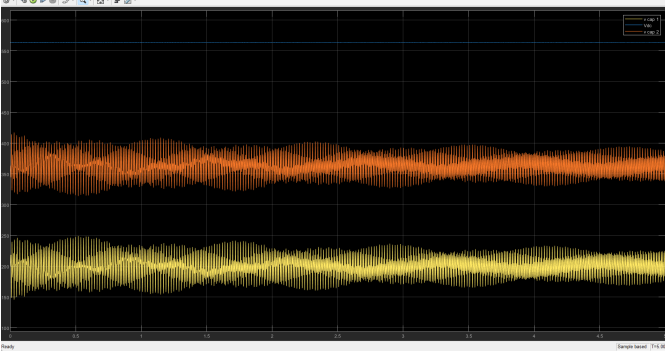


Fig. 16: DC bus capacitor voltages when simulated with an initial offset, in the presence of a 3-wire load ($V_{DC} = 230\sqrt{6}$, $V_{c1}(0) = 200$, $V_{c2}(0) = 230\sqrt{6} - 200$)

B. Forcing DC current through the mid-point

A common-cathode three-pulse diode based rectifier (3-phase) supplying an RL load can be used to introduce a DC component into the neutral point current [3]. This causes significant imbalance in the capacitor voltages, as seen in Fig.17.

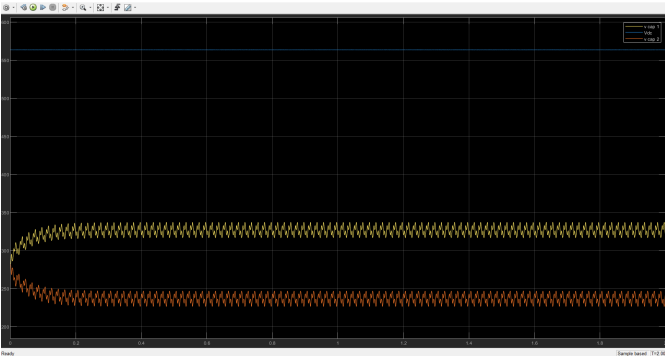


Fig. 17: DC bus capacitor voltages when a DC current is forced through the neutral point

VIII. SOLVING THE IMBALANCE PROBLEM

There are many approaches to correcting the DC bus capacitor voltage imbalance problem, based on different principles. A few of them will be looked into in the below section.

A. Adding an offset in SPWM

1) *Mathematical analysis:* For a balanced 3-phase load, like in Fig.13, if a bias is added to all 3 modulating waves, we would have [5]

$$v_{c,a} = m_a \sin(\omega t) + v_b$$

$$v_{c,b} = m_a \sin(\omega t - 120^\circ) + v_b$$

$$v_{c,c} = m_a \sin(\omega t + 120^\circ) + v_b$$

Plugging in the load currents and new duty cycles in equation (2), we have

$$i_o = \frac{m_a I_m}{2} [-\cos \phi - 2 \cos(2\omega t - \phi - 240^\circ)]$$

$$+ 2v_b I_m \sin(\omega t - \phi - 120^\circ)$$

$$; -\delta < \omega t < 60^\circ + \delta$$

$$i_o = \frac{m_a I_m}{2} [\cos \phi + 2 \cos(2\omega t - \phi)]$$

$$- 2v_b I_m \sin(\omega t - \phi)$$

$$; 60^\circ + \delta < \omega t < 120^\circ - \delta$$

$$i_o = \frac{m_a I_m}{2} [-\cos \phi - 2 \cos(2\omega t - \phi + 240^\circ)]$$

$$+ 2v_b I_m \sin(\omega t - \phi + 120^\circ)$$

$$; 120^\circ - \delta < \omega t < 180^\circ + \delta$$

$$i_o = \frac{m_a I_m}{2} [\cos \phi + 2 \cos(2\omega t - \phi - 240^\circ)]$$

$$- 2v_b I_m \sin(\omega t - \phi - 120^\circ)$$

$$; 180^\circ + \delta < \omega t < 240^\circ - \delta$$

$$i_o = \frac{m_a I_m}{2} [-\cos \phi - 2 \cos(2\omega t - \phi)]$$

$$+ 2v_b I_m \sin(\omega t - \phi)$$

$$; 240^\circ - \delta < \omega t < 300^\circ + \delta$$

$$i_o = \frac{m_a I_m}{2} [\cos \phi + 2 \cos(2\omega t - \phi + 240^\circ)]$$

$$- 2v_b I_m \sin(\omega t - \phi + 120^\circ)$$

$$; 300^\circ + \delta < \omega t < 360^\circ - \delta$$

$$\text{where } \delta = \sin^{-1}\left(\frac{v_b}{m_a}\right)$$

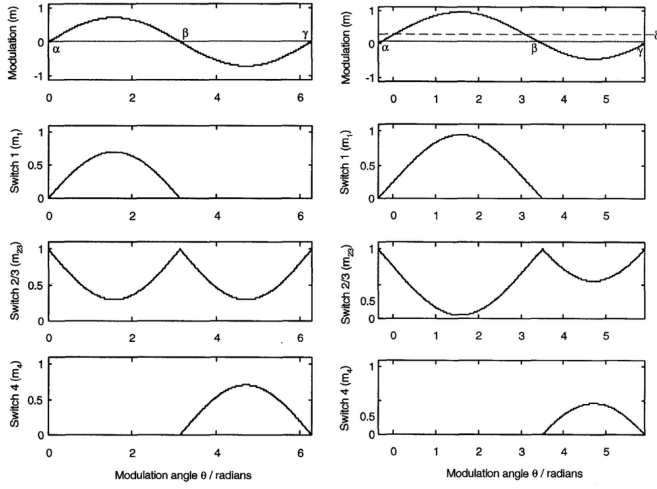
The integration of this neutral-point current gives a non zero value

$$\int_{-\delta}^{2\pi-\delta} i_o d(\omega t) = m_a I_m \cos(\phi) [-6\delta - 3\sin(2\delta)] \quad (3)$$

This DC value of current flowing through the neutral point gives a variation in the neutral point potential, according to equation (1).

2) *Alternate method for visualization:* The change in the fraction of ON time (duty ratio) can also be visualized using the diagram Fig.18 [6].

We clearly see that adding the offset upsets the symmetric distribution of duty ratios, which in turn leads to the different components of the neutral point current not cancelling out, causing a non-zero neutral point voltage.

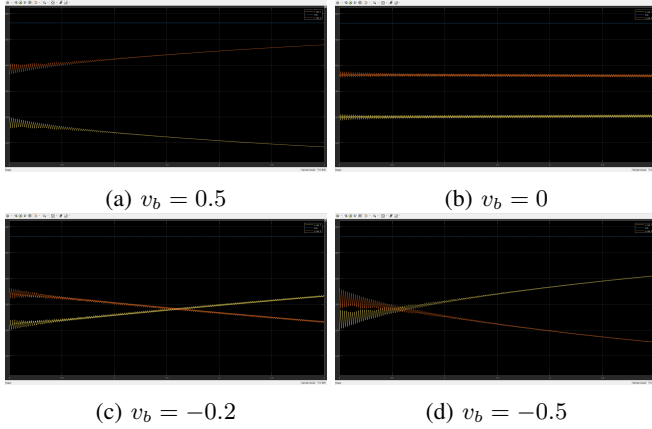


(a) Duty ratio vs ωt without any offset (b) Duty ratio vs ωt with the applied offset

Fig. 18: SPWM modulating wave and duty ratios of the 4 switches of a leg of a 3-level inverter

3) *Simulations:* Fig.19 shows the simulation results of adding such an offset to the modulating signal.

A suitable control scheme can be devised using the



(a) $v_b = 0.5$

(b) $v_b = 0$

(c) $v_b = -0.2$

(d) $v_b = -0.5$

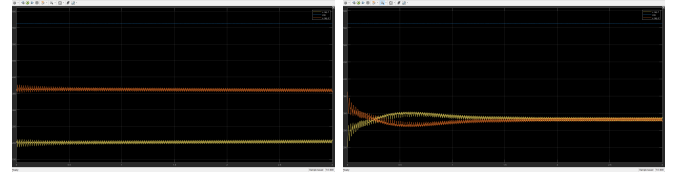
Fig. 19: DC bus capacitor voltages for different offsets v_b for $m_a = 0.5$

following principle, by continuously monitoring the drift of the capacitor voltages away from the DC bus midpoint and adding a suitable bias to the SPWM modulating signals to counter the imbalance. The error voltage is taken to be the difference in capacitor voltages and the offset is calculated by normalizing the error based on the DC bus voltage such that overmodulation does not occur.

The performance of a simple P controller with a suitable proportional constant has been shown in Fig. 20.

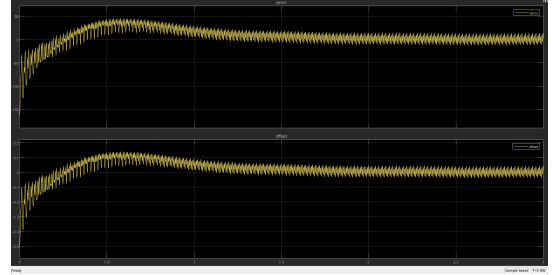
4) *Remarks:* Equation (3) and Fig. 19 show that -

- A positive offset causes a negative DC current at the neutral-point (and vice versa).



(a) Without any controller

(b) With control active



(c) Error voltage and calculated offset for $m_a = 0.5$

Fig. 20: Balancing using a P-controller that adds offsets to SPWM modulating waves

- The rate of change of the neutral-point voltage is higher for a higher magnitude of the offset v_b

It is essential to note that while adding such an offset may affect the phase-neutral voltages, this effect simply gets cancelled out in the line-line voltages, since all it does to the phase-neutral voltage is add a DC component. Therefore, such a balancing strategy does not affect the output in any way, since the fundamental component of the line-line voltages, which is what the load sees (after the filter) remains the same irrespective of the bias.

B. Usage of redundant switching states in SVPWM

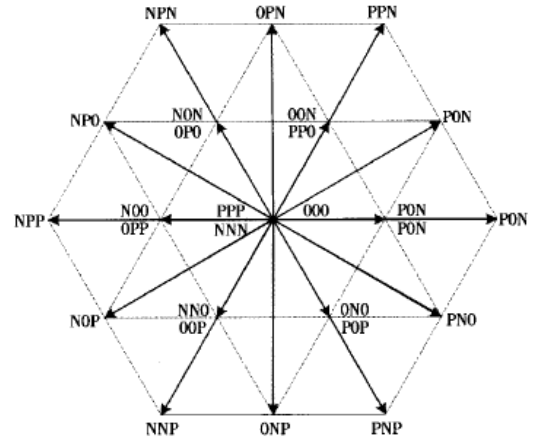


Fig. 21: State (phasor) diagram of SVPWM for 3-level inverters

As seen in Fig. 21, there are many redundant states that give the same space vector. That is, the line-line voltage output seen by the load is the same for any two redundant states.

Consider any two redundant states, say $(++0)$ and $(00-)$. We

notice that although they may provide the same output voltage, they result in the opposite direction of neutral point currents.

That is, $(++0)$ results in a current of i_b and $(00-)$ gives a current of $i_r + i_y = -i_b$ (for 3-wire load) at the neutral point. Depending on the direction of the instantaneous current in the b-phase, one state would inject current into the neutral-point, which would charge the bottom capacitor and charge the top capacitor, while the other state would pull current out of the neutral point, charging the top capacitor and discharging the bottom one.

Neutral point control can be implemented through selection of the switching state used for a particular voltage vector [6]. That is, the fraction of ON time given to either redundant state could be manipulated to change the current direction in order to correct the neutral point imbalance for a given switching cycle.

C. External circuits for charge transfer

The above methods are effective only for correcting minor imbalances, for instance, those caused due to non-idealities, especially in a system having a 3-wire load. They have a limited region of operation; for example to avoid overmodulation in SPWM and to keep the time fractions positive in SVM, due to which they may not be able to completely bring the imbalance down to zero in the case of a stronger imbalance (like when DC current flows through the neutral point). The introduction of a 4th wire may also affect the output voltage waveform in the case of SPWM, since there is opportunity for the introduction of common mode components.

An effective way to solve such a severe imbalance would be to simply transfer the charge of the capacitor with the higher voltage to the capacitor with lower charge using an intermediate inductor for temporary storage of charges. Many forms of circuits that use the same principle have been proposed, with the main difference being the switches used ([7] and [3]).

1) *Chopper circuit for balancing DC bus capacitor voltages:* Refer Fig. 22

2) *Working principle of the chopper circuit:* As an example, if it is assumed that there is a positive dc component in the neutral current, it is evident that V_{dc2} will become greater than V_{dc1} over time. Hence, in order to equalize the voltages, the switch S_n (refer Fig.22) is operated so that energy is transferred from the bottom capacitor to L_{ch} . Once the switch S_n is turned off, the energy in the inductor is transferred to the top capacitor through the antiparallel diode D_p . Similarly, for a negative dc component, V_{dc1} will become greater than V_{dc2} , requiring the operation of switch S_p and diode D_n .

3) *A control strategy for the chopper switches:* The error voltage is obtained by subtracting the desired reference value from the voltage of the capacitor which experiences a boost in the voltage due to the dc component. This ensures that the controller always acts on a positive error. This positive error is then compared with a positive triangular carrier, to generate the switching signals for the chopper switch corresponding

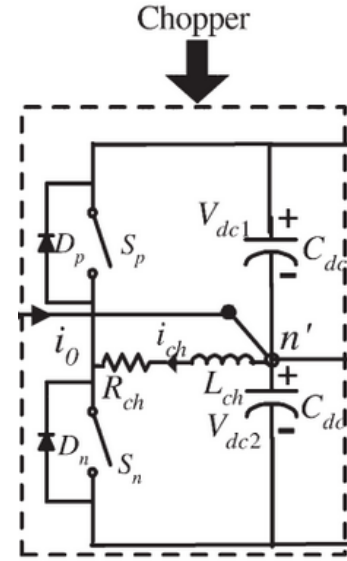


Fig. 22: Chopper circuit connected to the DC bus capacitors

to the capacitor whose voltage increases. The unipolar carrier ensures that the operation of the chopper is stopped when the error becomes zero or negative.

Whenever the error voltage is higher than the carrier, the duty cycle of operation is automatically adjusted, depending on the magnitude of error. If the error is high, the duty ratio also correspondingly increases, so that the excess charges are discharged from the capacitor, transferring the energy to the inductor. Once the corresponding switch is off, the inductor current starts freewheeling through the antiparallel diode of the other switch, hence transferring the excess energy to the voltage-deficit capacitor.

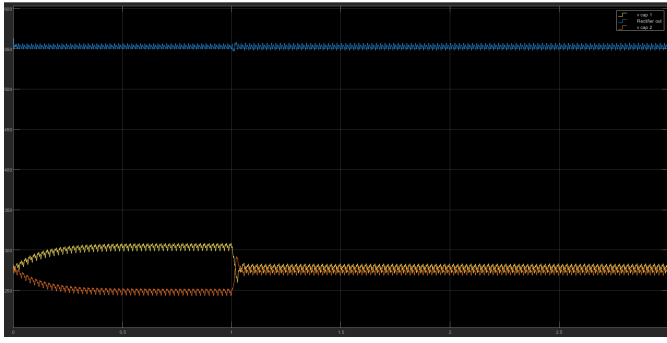
4) *Simulation of the chopper action:* From Fig. 23, it is evident that the capacitor voltages are forced to the desired reference value (here $V_{DC}/2$) when the chopper is switched ON (allowed to operate).

IX. CHOICE OF BALANCING STRATEGY

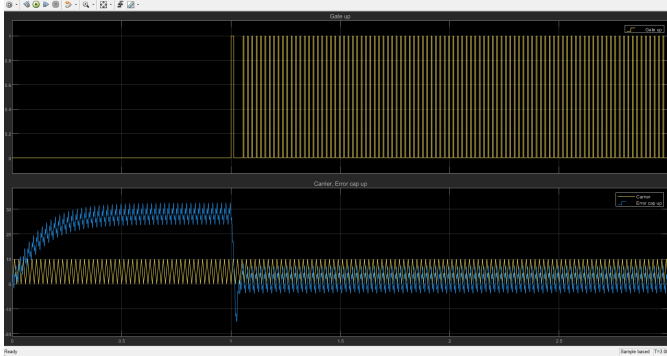
Depending on the type of load, allowable power consumption and other inverter parameters (like DC bus utilization or maximum THD in the pulsed output), the choice of the optimal strategy to correct the imbalance could vary.

A. External charge balancing (chopper) circuit

While the chopper is a prime choice in a system with a 4-wire load, there are certain downsides that come as part of this circuit. The use of an inductor makes the hardware a lot costlier and bulkier. Extra losses are also seen due to the addition of the 4 switches and the passive components. Thus, in the case of minor imbalances, such as those caused due to non-idealities in a system having a 3-wire load, the techniques that take advantage of the switching schemes are preferred.



(a) Chopper on at $t = 1\text{s}$



(b) Error voltage and duty cycle of switch

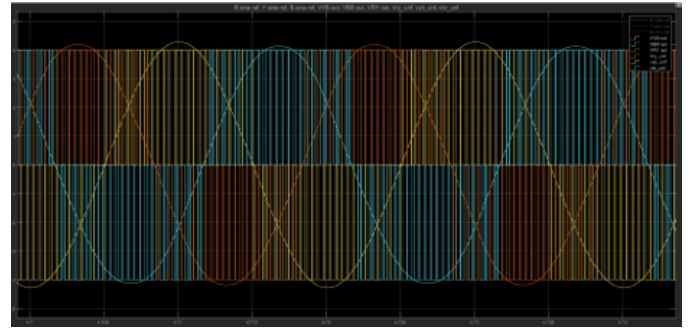
Fig. 23: DC bus capacitor voltages when subjected to balancing by the chopper

B. A comparison between SVM and SPWM

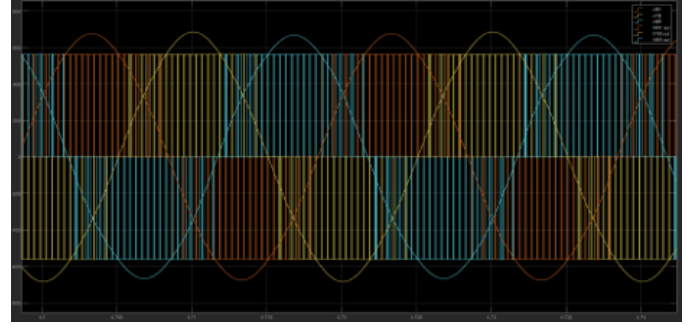
SPWM and SVM are both leading modulation strategies for the operation of inverters, with both of them delivering nearly equal performances in both 2 and 3-level inverters. While SPWM is much easier to achieve and has both digital and analog implementations (SVM needs a Digital Signal Processor to be carried out), SVM has a slight edge over SPWM in terms of performance. As seen in Fig. 24 and 25, SVM has a 15% better DC bus utilization compared to SPWM, apart from giving a lower THD.

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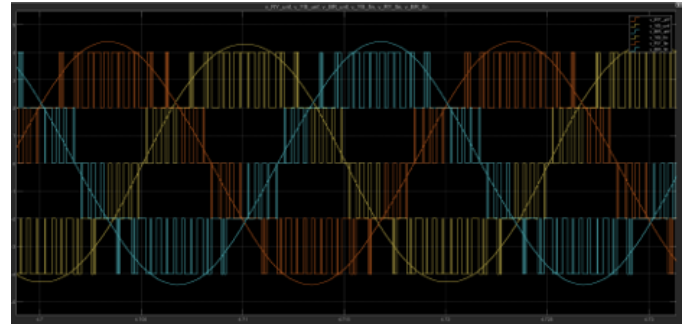


(a) SPWM: THD = 66.41%

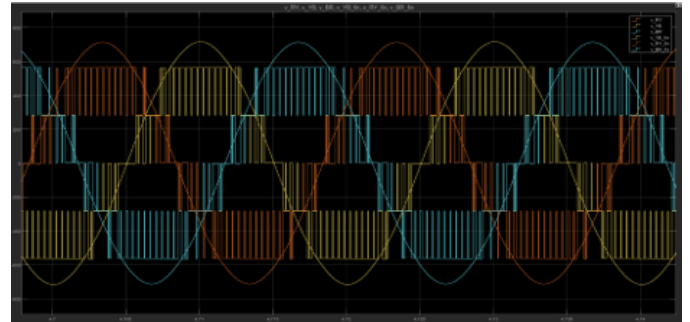


(b) SVM: THD = 60.51%

Fig. 24: Output line-line voltage of a 2-level inverter



(a) SPWM: THD = 35.81%



(b) SVM: THD = 28.96%

Fig. 25: Output line-line voltage of a 3-level inverter

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